van Emde Boas trees exercises

20-E01 Bitmap successor

Difficulty: Basic

Bitmap B=00100110 has positions increasing rightward from leftmost0. Find successor2,predecessor6,and min.

20-E02 Cluster index

Difficulty: Advanced

u=256,upper=lower=16. Find cluster and local value for key173.

20-E03 vEB recursion depth

Difficulty: Research

For $u = 2^{32}$,estimate levels down to universe2 when $\log_2 u$ halves each time.

20-E04 Updates with min excluded from clusters

Difficulty: Advanced

At u=16,S={2,5,6,9,12,15},give min/max,four local cluster sets,and summary. Trace insert1,delete1,delete15,including promotion,removal,and summary changes.

20-E05 Two calls need not mean two expensive recursions

Difficulty: Research

Insertion into an empty group updates summary and cluster; deletion of a singleton group also updates both. Why is the recurrence not2T($\sqrt{}$u)+O(1)? Compare and solve uncached proto-minimum/proto-successor recurrences.

20-E06 Large-universe initialization must be counted

Difficulty: Research

For preallocated u=2^32 and only n=1000 stored keys,why may a comparison tree be better? Each base node uses at least one32-bit word. Count only the u/2 cluster-derived bases to give a space lower bound,and compare total time for m operations.

> Complete your derivations before opening the separate solutions PDF.

van Emde Boas trees exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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